ODA CANVAS · IMPLEMENTATION VIEW

ibn-core as an ODA Canvas Component

How the Apache-2.0 ibn-core framework packages as a first-class **TM Forum ODA Canvas Component** (v1alpha3) — registering its TMF921 API, declaring its Canvas use-case bindings, and forward-declaring its AI-native capabilities (LLMs, MCP, A2A) for an AI-Native Canvas.

TM Forum ODA Canvas · `oda.tmforum.org/v1alpha3`

Helm chart `ibn-core 2.2.0` · RFC 9315 · TMF921 v5.0.0

↔ Customer view · ↔ Red-blue view · ⬇ **Download (HTML)**

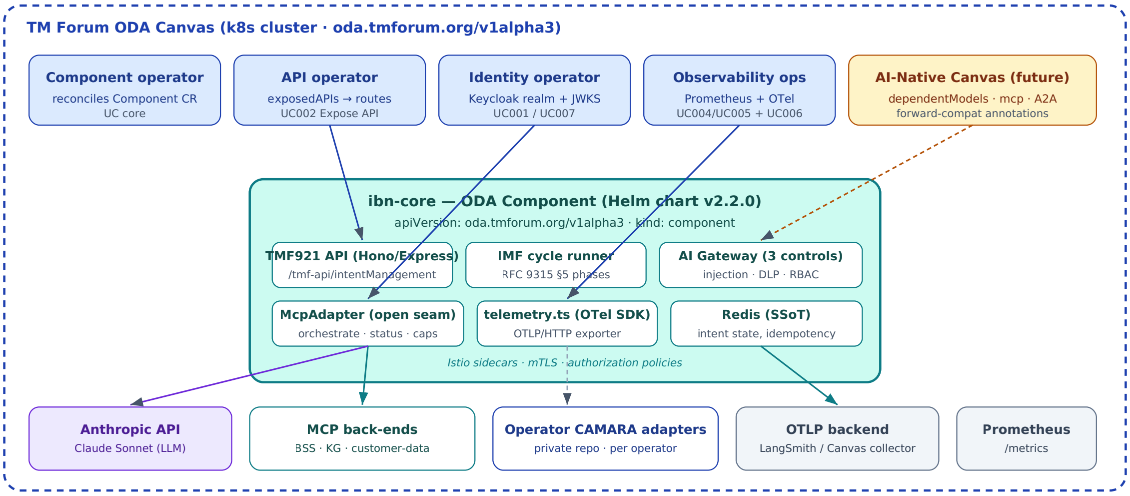
1 · Executive summary

ibn-core does not need a Canvas to run, but when deployed into one it surfaces as a typed, observable, governed component — and is forward-compatible with the AI-Native Canvas design.

v1alpha3 ODA Component CR registered by the chart	**5+** Canvas UCs the chart binds to: UC001 · UC002 · UC004/5 · UC006 · UC007	**4** AI-Native forward-compat annotations (`dependentModels`, `mcpInterfaces`, `agentCard`, `evaluationDataset`)

The Canvas value proposition for ibn-core: identity bootstrap is delegated to a Canvas identity operator (Keycloak realm), API exposure to a Canvas API operator (TMF921 routes), classic and custom observability to Canvas observability operators (Prometheus + OTel/Canvas collector), and — when the AI-Native Canvas extensions land — the LLM dependency, MCP tool surface, and A2A agent card are discoverable via the same Component CR rather than via out-of-band metadata.

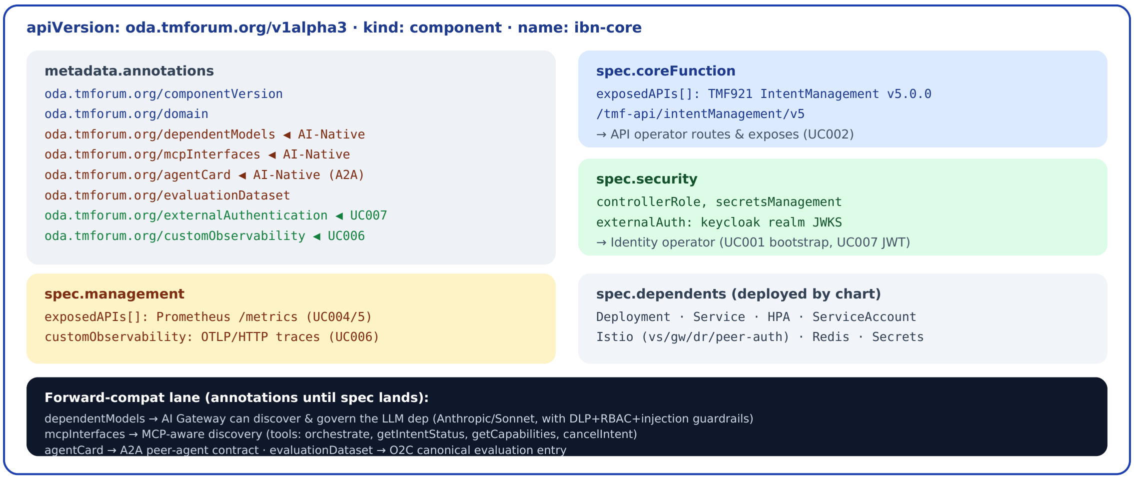
2 · Where ibn-core sits in the Canvas (C4 context)



The Component CR is the contract. Canvas operators (top row) reconcile it; ibn-core (centre) provides the implementation; external systems (bottom row) sit outside the Canvas boundary or in a private operator-adapter repo.

3 · ODA Component anatomy — the contract

The Helm chart renders one `oda.tmforum.org/v1alpha3 · kind: component` resource that declares everything Canvas operators need to bind to ibn-core.



Component CR layout. Blue annotations are standard v1alpha3; amber are AI-Native forward-compat; green are the Canvas UC bindings UC006/UC007 surface as annotations until spec fields exist.

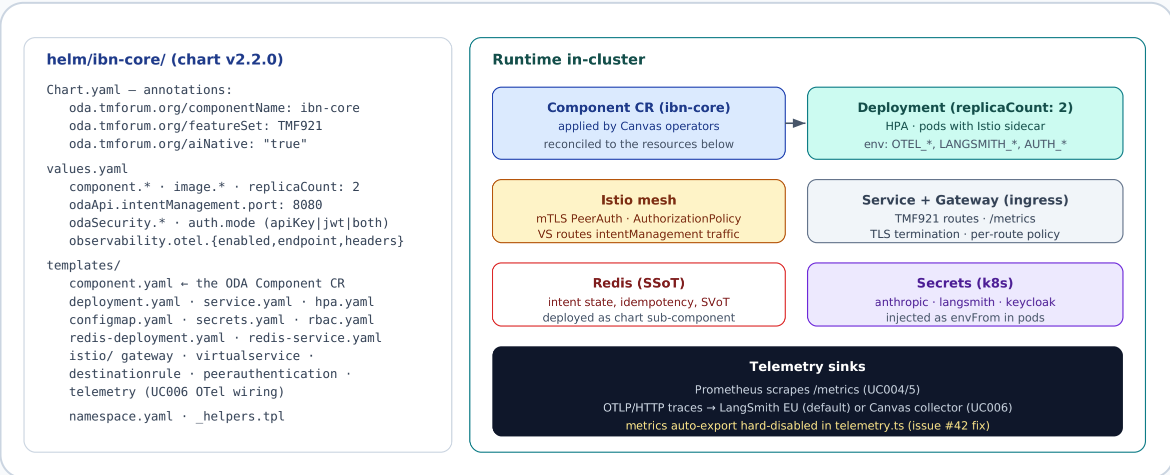
4 · Canvas use-case coverage

The Helm chart declares — and the running component implements — the following Canvas use cases. UC006 and UC007 were the headline deliveries in `v2.2.0/v2.3.0`.

UC	Canvas use case	How ibn-core binds	Status	Evidence
UC001	Identity bootstrap	Component declares `security.controllerRole`; chart provisions the ServiceAccount + Keycloak realm-binding inputs.	implemented	`templates/component.ya` `templates/rbac.yaml`
UC002	Expose API	`coreFunction.exposedAPIs[]` registers the TMF921 v5.0.0 IntentManagement API; Canvas API operator publishes routes.	implemented · 83/83 CTK	`component.yaml` · `templates/istio/virtualservice.yaml`
UC004 / UC005	Classic observability (metrics / logs)	`management.exposedAPIs` publishes Prometheus `/metrics` on port 9090; pino structured logs.	implemented	`src/metrics.ts` · `helm/` `templates/service.yaml`
UC006	Custom observability (OTel traces)	`customObservability` annotation + `src/telemetry.ts` ships OTLP/HTTP traces (LangSmith default; Canvas-collector overridable).	Phase 1 merged (#35)	UC006_CANVAS_CTK_RES
UC007	External authentication	`externalAuthentication` annotation; `src/auth-jwt.ts` validates RS256 against a Keycloak realm JWKS.	implemented (#34)	`src/auth-router.ts` · `s` `auth-jwt.ts`
—	AI-Native (dependent models / MCP / A2A)	Forward-compat annotations declare LLM dep, MCP tool surface, A2A agent card, evaluation dataset.	annotation-level	component.yaml annotations Native Canvas design

UC	Canvas use case	How ibn-core binds	Status	Evidence
UC008+	Compliance / lifecycle / federation	Roadmap.	planned	—

5 · Deployment topology & operations



Left: the Helm chart files. Right: the in-cluster shape the Canvas operators produce when they reconcile the Component CR. The chart is operator-installable today; AI-Native-specific operators (when they land) will additionally consume the annotation forward-compat lane.

6 · AI-Native Canvas — forward-compat lane

Until the AI-Native Canvas spec formalises these fields, ibn-core declares them as annotations on the Component CR. When the spec lands, they migrate to first-class spec fields without breaking deployments.

`dependentModels` · LLM declaration

Declares the external LLM dependency (Anthropic Claude Sonnet) and the guardrails in front of it. Lets a Canvas AI Gateway discover and govern model usage at the component level rather than via out-of-band documentation.

```
- provider: anthropic
  model: claude-sonnet-4-…
```

`mcpInterfaces` · MC

Lists the tools the McpAd
seam to operator adapter
aware to Canvas-side too

```
- name: intentOrc
  protocol: mcp ·
```

```
    useCase: "RFC 9315 NL intent translation (§5.1.2)"
    guardrails: { dlp, toolPolicy, promptInjection }
```

```
    tools: [orchest
      getCapa
```

`agentCard` · **A2A peer contract**

Identifies ibn-core as an A2A agent with named capabilities and an endpoint — so peer agents can discover and invoke it through Canvas-mediated agent-to-agent flows.

```
name: ibn-core Intent Agent
capabilities: [intentIngestion, intentTranslation,
               intentOrchestration, complianceMonitoring]
a2aEndpoint: /api/v1/intent/probe
```

`evaluationDataset`

Points Canvas evaluation
to-Cash evaluation set —
deterministic AI behaviou

```
oda.tmforum.org/e
  "docs/complianc
```

7 · Evidence & honest caveats

What's verified today. Helm chart v2.2.0 packages and applies as a Canvas Component CR on any cluster with the `oda.tmforum.org/v1alpha3` CRD installed (via the canvas-oda umbrella chart). UC006 ingest path verified end-to-end against LangSmith EU. UC007 JWT validation against a Keycloak realm JWKS verified in #34. UC004/5 metrics and UC002 API exposure pass their respective CTKs.

What's still pending.

- **UC006 CTK against a Canvas-managed collector** — current Phase-1 evidence is against LangSmith EU. The CTK run table in `UC006_CANVAS_CTK_RESULTS.md` §4 is filled in once a Canvas with a managed OTel collector becomes reachable from our environment.
- **AI-Native Canvas spec fields** are not yet first-class — the dependentModels / mcpInterfaces / agentCard / evaluationDataset annotations migrate to spec when TM Forum publishes the schema. No deployment break expected.

- **CI status checks** on `main` are billing-locked at the GitHub account level; merges proceed via a temporary `enforce_admins` toggle (audited, immediately restored).

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ibn-core — ODA Canvas implementation view. References: TM Forum ODA Canvas, Component v1alpha3, AI-Native Canvas design (forward-compat), TMF921 v5.0.0, RFC 9315. Standalone document; all diagrams inline SVG.